Requirements Specification for an AI-Assisted Guest Support Assistant

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This form is completed in accordance with the Software Engineering Body of Knowledge (SWEBOK).

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1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the Phase 1 AI-Assisted Guest Support Assistant for Vacations for YOU. This Software Requirements Specification (SRS) serves as a reference for stakeholders, including project sponsors, instructors, and development team members, by describing system scope, capabilities, constraints, and expectations. The document provides a foundation for system design, implementation, and evaluation during the capstone project.

1.2 Scope

The document specifies the requirements for the Phase 1 proof of concept AI-assisted guest support system for Vacations for YOU. The system outlined is designed to support guests with existing reservations by providing a fully conversational chatbot experience that delivers automated responses to common post-booking questions through a web-based chat interface.

The Phase 1 system will employ a large language model (LLM) to generate responses based on read only reservation, property, and policy data from the StreamlineVRS property management platform or an approved sandbox environment. The AI assistant will provide informative responses and present clear escalation guidance when questions fall outside of the scope or confidence threshold.

The project excludes booking creation or sales, payment processing, service order placement including but not limited to maintenance, account creation, or authentication beyond what is necessary to provide reservation information. The project is intentionally structured to validate feasibility, usability, and call-deflection potential for future channel expansion.

1.3 Definitions, Acronyms, and Abbreviations

- **AI** – Artificial Intelligence
- **UI** – User Interface
- **LLM** – Large Language Model
- **POC** – Proof of Concept
- **SRS** – Software Requirements Specification

- **StreamlineVRS** – A property management system used by Vacations for YOU to manage reservations and property information
- **Guest** – An individual with an existing reservation at a Vacations for YOU property
- **Call Deflection** – Reduction in customer support calls through automated self-service solutions

1.4 Overview of the Document

This document is organized into five main sections. Section 1 introduces the purpose, scope, and structure of the requirements of specification. Section 2 provides an overall description of the system, including its intended users, operating environment, and constraints. Section 3 describes the system, user, hardware, and software interfaces. Section 4 defines the functional and non-functional requirements of the system. Section 5 outlines key design considerations that influence implementation while remaining independent of specific technologies or frameworks.

1.5 References

IEEE. ISO/IEC/IEEE 29148: Systems and Software Engineering — Life Cycle Processes — Requirements Engineering. IEEE Standards Association, 2018.

Vacations for YOU. Capstone Project Specification: Reservation Support Chat Platform (Phase 1). Streamline Vacation Rental Software. Open API Documentation. https://www.streamlinevrs.com/features/open-api/

1.6 Assumptions

Vacations for YOU utilize the StreamlineVRS platform for reservation and property management. The project assumes that the sponsor will provide either read-only access to the StreamlineVRS API or appropriate sandbox API for documentation for development and testing purposes. If direct access to live systems is restricted, the sponsor may supply sample or anonymized data representative of production data structures.

The system shall be designed to operate with mock or simulated data when necessary and support seamless replacement of mock integrations with live StreamlineVRS API connections in future phases. The architecture should allow real API integration to be introduced with minimal refactoring to core system logic.

1.7 Constraints

The development and evaluation of this system are subject to the following constraints. The project must be completed within the duration of a single academic semester. No live customer or order data may be accessed, stored, or processed at any time. System integration with sponsor platforms shall be limited to read-only access, whether through live endpoints or approved sandbox environments.

Finacial and payment information will not be included as well as the ability to alter reservations such as booking, cancelling, or adjusting reservations.

All changes to the defined project scope must be formally approved by the sponsor and documented prior to implementation. These constraints are intended to reduce risk, protect sensitive data, and ensure that the project remains aligned with sponsor expectations and academic requirements.

1.8 Explicit Exclusions

The following topics are out of scope for the Chatbot Assistant:

- Reservation creation
- Reservation modification
- Cancellation processing
- Payment handling
- Financial balance inquiries
- Refund processing
- Loyalty account modification
- Account authentication beyond basic validation

1.9 Questions

Date	Questions
2/4/2026	1. What is the direction this project is intended to go once there is proof of concept? 2. What response time target should we design? 3. What sort of data can we expect to be utilized? For instance, would the chatbot be able to give suggestions?

	3.1. Sandbox account for testing 4. Do we need a server for this project? 5. Are there required security/compliance standards we must follow (e.g., data minimization, logging restrictions, encryption in transit/at rest, authentication requirements, least-privilege access)?
2/18/2026	1. How do Memberships impact the chatbots responses? 2. What would be a fair cut off time for a conversation to be considered over? 3. How should the chatbot respond to questions outside of scope? 4. What happens if authentication fails? 5. Should conversations persist across sessions? 6. Does the Chatbot handle complaints?

Table 1. Questions log

2. Overall Description

2.1 Product Perspective

The Vacations for YOU AI Guest Support Platform is being developed as a proof-of-concept solution to address the high volume of routine customer inquiries currently overwhelming the call center at a vacation rental business. The system will integrate with the existing StreamlineVRS API, which manages property information and reservation data across the company's rental portfolio. Right now, call center agents spend a lot of time answering the same basic questions about check-in times, property amenities, and company policies, so this platform aims to automate those repetitive interactions by using AI to understand guest questions and pull the correct information directly from StreamlineVRS. The system sits between the guest and existing vacation rental management infrastructure, querying the StreamlineVRS API for real data instead of making things up, which should keep responses accurate and grounded in actual property information. For questions that are too complicated or outside the AI's scope, the system will escalate to a human agent. This is designed as a proof-of-concept focused on core functionality that demonstrates feasibility rather than building a fully production-ready system.

2.2 Product Functions

The system will provide a conversational chat interface that accepts natural language questions from guests and maintains conversation context across multiple exchanges. Core functions include querying the StreamlineVRS API for property information, accessing reservation data, and answering policy questions about cancellations, pets, parking and house rules. The platform will use prompt engineering techniques to ground responses in actual API data and minimize hallucinations, recognize when questions exceed the AI's capabilities and escalate to human agents with conversation context, and handle inquiries across the entire property portfolio while differentiating between individual properties.

2.3 User Classes and Characteristics

The primary users are guests who have existing reservations only, with varying levels of technical comfort accessing the chat from phones, tablets, or computers, often with time-sensitive questions about check-in times, pet policies, parking, amenities and key pickup procedures. Secondary users are call center agents who handle escalated cases and need to see conversation history to understand what the guest has already asked without making them repeat information. Tertiary users are system administrators who monitor performance, review logs for errors or problematic interactions, and adjust prompts or system behavior based on analytics about query types, escalation rates and system accuracy.

2.4 Design and Implementation Constraints

The system must work within the limitations of the StreamlineVRS API. We need to be mindful of API costs since OpenAI and Azure OpenAI charge per token, implement proper authentication before showing reservation data to protect guest privacy, and ensure the AI cannot modify or cancel reservations or handle financial transactions. The system needs to handle multiple concurrent users, maintain high accuracy, and work within the scope of limitations.

2.4.1 Supported Vs Unsupported Question Taxonomy

Supported	Unsupported
• Reservation status inquiries	• Booking Requests • Pricing and financial concerns

• Property information (amenities, dates, property information) • Policy questions • Check-in/check-out time questions	• Legal advice/Medical advice • Questions unrelated to reservations/properties • Requests to override system policies

Table 2. Question Taxonomy (Initial)

2.5 Assumptions and Dependencies

We're assuming that the StreamlineVRS API is stable and well-documented with endpoints providing property details, amenities, policies, and reservation information, that most guest questions are informational rather than transactional, that guests will be willing to interact with a chatbot for basic questions especially outside business hours, that OpenAI or Azure OpenAI will remain available with consistent API access and pricing during development, and that queries will be in English for this proof-of-concept. The system depends on the StreamlineVRS API being available and providing accurate data, the LLM provider's service uptime and API stability, sponsor access for API credentials and business process clarification, hosting infrastructure for deploying the backend and frontend, the Python ecosystem, the chosen frontend framework, and reliable internet connectivity for both the system and users to enable API calls to StreamlineVRS and the LLM provider.

3. System Interfaces

3.1 System Interface

The system interfaces with external services to retrieve reservation, property, and policy data and to generate AI-assisted responses. Primary system integrations include the StreamlineVRS property management platform, which provides read-only access to reservation and property metadata, and a large language model (LLM) service used to generate conversational responses. All system interactions are performed through secure API-based communication.

3.2 User Interface

The system provides a web-based chat interface accessible through standard desktop and mobile web browsers. The interface supports real-time, conversational exchanges between the guest and the chatbot. It provides clear transition to human support when required. The interface allows guests to validate an existing reservation, submit natural language questions, view system responses, and receive escalation guidance when applicable. The user interface is designed to support accessibility, clarity, and ease of use without requiring prior training.

3.3 Hardware Interfaces

The system does not require any specialized hardware interfaces. It is intended to operate on standard cloud or server infrastructure and be accessed by users through common consumer devices such as smartphones, tablets, and personal computers with internet connectivity.

3.4 Software Interfaces

The system interfaces with external software services, including the StreamlineVRS API for reservation and property data access and an external LLM service for natural language processing. The system is designed to operate within a standard web application environment and relies on commonly available web technologies for communication and data exchange. Specific implementation technologies are not mandated by this specification.

4. Requirements

4.1 Functional Requirements

FR-01: Simple Entry Point

The system shall provide a simple web-based entry point that allows guests to access the AI-assisted support from the current interface without needing to find a different web location or install additional software.

FR-02: Reservation Validation

The system shall allow guests to validate an existing reservation using a non–PII-heavy identifier such as a reservation number.

FR-03: Conversational UI

The system shall provide a conversational UI so that guests can use natural language questions and receive responses with optimized clarity and speed.

FR-04: Human Support Instructions

The system shall provide clear instructions for contacting human support when an inquiry cannot be solved or is given a pre-defined case where the solution must be escalated, by the AI assistant.

FR-05: Membership Recognition

The system shall retrieve and recognize guest membership or loyalty tier status when available. The system shall tailor responses based on sponsor-defined membership policies when applicable.

FR -06: Transfer Confidence

The system shall provide a mechanism to transfer an active conversation to a human support agent when a question cannot be confidently resolved by the chatbot.

FR –07: Intent Classification

The system shall classify guest questions or concerns into common post-booking categories.

FR –08: User Context-aware Responses

The system shall generate responses that are context-aware by using reservation and property data associated with the guest's validated reservation.

FR –09: Hallucination Minimization

The system shall constrain AI-generated responses to retrieved reservation, property, and policy data to minimize hallucinations.

FR -10: Authoritative Data Ground

The system shall generate responses grounded exclusively in retrieved or sponsor-approved data sources.

FR-11: Confidence Evaluation

The system shall evaluate response confidence prior to delivering a response to the guest.

FR-12: Escalation Triggering

The system shall transfer the conversation to human support when confidence thresholds are not met.

FR –13: Response Justification

The system shall provide supporting references or justification when required by sponsor-defined rules.

FR –14: Conversation logging

The system shall log quest guests and concerns, the system responses, and escalation events for evaluation and analysis purposes.

FR-15: Write Confirmation Requirement

The system shall require explicit guest confirmation before executing any operation, specifically written. This identifier will be a phone number and/or reservation number.

FR –16: Conversational Context Handling

The system shall support multi-turn conversational interactions, allowing the chatbot to maintain context across sequential guest messages within a single chat session.

FR-17: Session Timeout Handling

The system shall automatically terminate inactive sessions after a predefined timeout period.

FR –18: Save conversations for Agent referral

The system will save the chat so that it can be attached to a call ticket and referred to later by an agent.

FR-19: Write Operation Logging

The system shall log all write operations including timestamp, session ID, and modified data fields. Each Conversation will have a unique ID associated with it.

FR-20: Financial Operation Restriction

The system shall not process payments, refunds, or financial transactions.

FR-21: Unauthorized Access Prevention

The system shall prevent access to reservation data without successful validation.

4.2 Non-Functional Requirements

These non-functional requirements define quality attributes that constrain how the functional requirements are implemented and evaluated.

NFR –01: Accuracy

The system shall achieve at least 90% factual correctness when evaluated against known reservation and property data during controlled testing.

NFR –02: Quick Response Time

The system shall return a response within 3 seconds for 95% of requests under normal operating conditions.

NFR –03: Reliability

The system shall provide consistent responses for identical questions when given the same reservation information or property information.

NFR –04: Usability

The system shall provide an intuitive user interface that allows guests to submit questions and receive responses without prior to website interaction or other training.

NFR –05: Maintainability

The system shall be designed so that prompts, policies, and data sources can be updated without requiring significant changes to core application logic.

NFR –06: Scalability

The system shall be able to support multiple chat sessions to support the proof of concept and later shall be able to be expanded to support other interfaces

NFR –07: Auditability

The system shall log interactions and system events in a structured format to support analysis, evaluation, and troubleshooting.

NFR –08 Security

All external API communications shall occur over encrypted HTTPS connections.

NFR-09: Hallucination Rate Control
The system shall maintain a hallucination rate below 5% during controlled evaluation testing.

NFR-10: Modular Architecture
The system shall be designed so that AI prompts, policy rules, and data source configurations can be updated without modification to core session management logic.

5. Design Considerations

Section 5, Design Considerations outlines the key design constraints and architectural principles that influence implementation decisions while remaining independent of specific technologies or frameworks. This approach preserves flexibility in future development and supports phased system expansion.

5.1 Software Design

The system should follow a modular architecture that separates the user interface, AI processing logic, data access layer, and external service integrations. Separation supports maintainability, testing, and future expansion, including the addition of alternative interaction channels such as voice-based support.

The software design should be channel-agnostic, ensuring that core AI reasoning and data retrieval capabilities are decoupled from specific interaction modalities. Phase 1 implementation will focus on a web-based chat interface to enable validation and verification of AI logic before integration with additional communication channels in future phases.

5.2 Database Design

The system should minimize persistent data storage and avoid duplication of reservation or property data managed by external systems. Any locally stored data should be limited

operational needs such as conversation logs, anonymized usage metrics, and system performance indicators. Guest-identifiable information should not be retained beyond the duration required for active chat session processing. All stored data should comply with applicable data protection and privacy best practices and align with existing organizational data protection procedures.

Conclusion

Summary

This document defines the requirements for a Phase 1 AI-assisted guest support system for Vacations for YOU. The specification establishes system scope, functional and non-functional requirements, interfaces, and design considerations necessary to support post-booking guest inquiries using a web-based conversational interface. By focusing on read-only integration with existing reservation and property data and emphasizing safe AI behavior with clear escalation paths, this requirements specification provides a foundation for implementation, evaluation, and future system expansion.

Appendix

Table 1. Questions log

Table 2. Question Taxonomy

Change Log

Version	Date	Description
1.0	January 2026	Initial SRS creation

2.0	February 2026	• Updated format
		• Added 1.8 Explicit Exclusions
		• Updated Requirements List:
		  ○ FR-5
		  ○ FR-10
		  ○ FR11-FR21
		  ○ NFR 8-NFR 10
		• Added Table 2. Question Taxonomy